# Brand.csv — Exploratory Data Analysis Report

This report documents the steps and visualizations from the exploratory data analysis (EDA) performed on the Brand dataset (influencer–brand metrics across Instagram, YouTube, and TikTok).

## EDA steps performed

### *1. Data loading and basic overview*

Loaded Brand.csv and inspected shape (rows × columns), column names, and data types. Reviewed first rows and ran descriptive statistics (e.g. describe()) on numeric columns.

### *2. Missing values analysis*

Computed missing counts and percentages per column. Identified columns with high missingness (e.g. TIKTOK_USER_ID, YOUTUBE_USER_ID, ACCOUNTNAME, INSTAGRAM_USER_ID) to inform cleaning and modeling.

### *3. Correlation analysis*

Built a correlation matrix for key metrics (followers, engagement, sales, orders, clicks, etc.) and visualized it as a heatmap to identify strong relationships and redundancy.

### *4. Distribution of Instagram followers*

Plotted the distribution of Instagram follower count. Used a log scale to handle right skew and visualize the spread of audience sizes.

### *5. Engagement rate by platform*

Compared distributions of engagement rate across Instagram, YouTube, and TikTok to understand platform-level engagement patterns.

### *6. Distribution of business metrics*

Examined distributions of total sales and total orders (on log scale) to understand conversion and revenue patterns.

### *7. Box plots for key metrics*

Created box plots for key numeric variables (on log scale where appropriate) to assess spread, central tendency, and outliers.

### *8. Scatter: followers vs sales*

Plotted Instagram follower count against total sales (log–log) and computed correlation to assess how reach relates to revenue.

### *9. Scatter: engagement rate vs orders*

Plotted Instagram engagement rate against total orders to explore whether engagement quality relates to conversions.

### *10. Top brands by total sales*

Aggregated total sales by ACCOUNTNAME and visualized the top 15 brands by revenue.

### *11. Top brands by engagement*

Aggregated Instagram total engagement by ACCOUNTNAME and visualized the top 15 brands by engagement.

### *12. Missing values visualization*

Created a bar chart of missing value counts per column for quick identification of data completeness.

### *13. Platform presence*

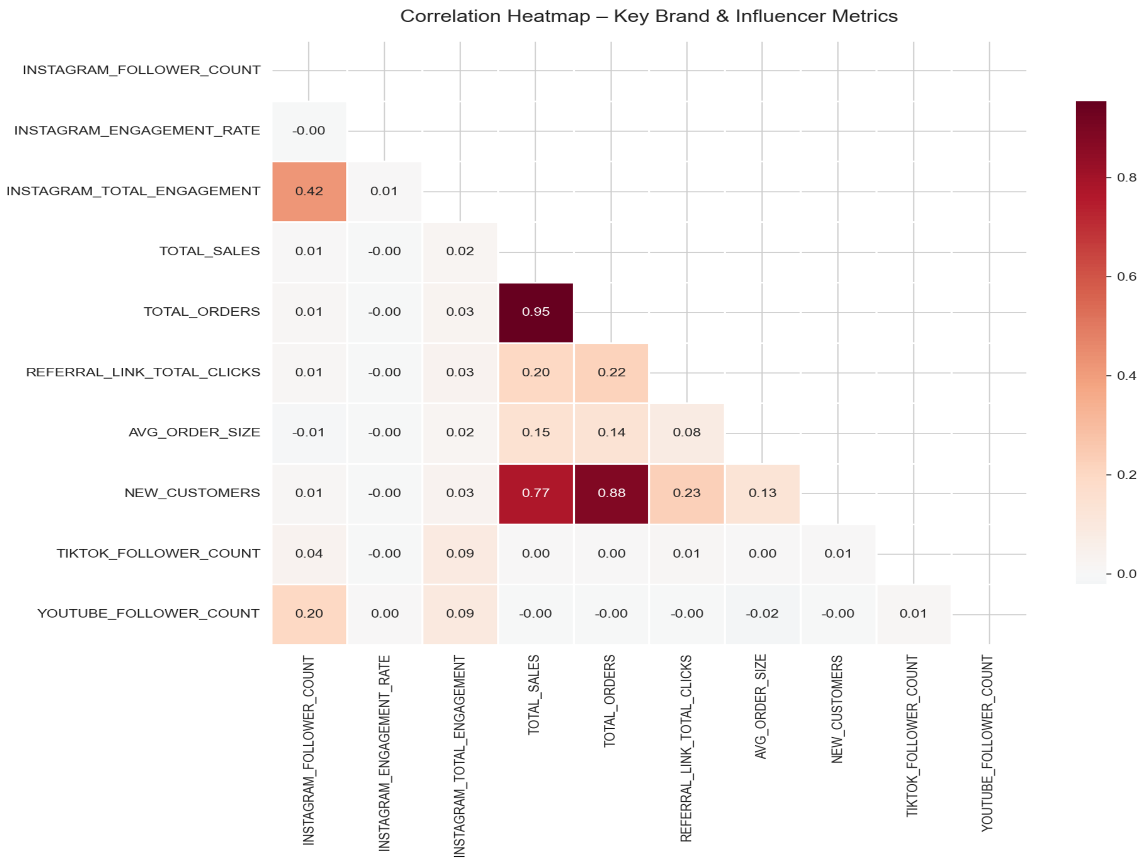
Computed and visualized how many records have non-zero follower counts on Instagram, YouTube, and TikTok to show multi-platform coverage.

### *14. Top brands by influencer count*

Counted influencer records per ACCOUNTNAME and visualized the top 15 brands by number of affiliated influencers.

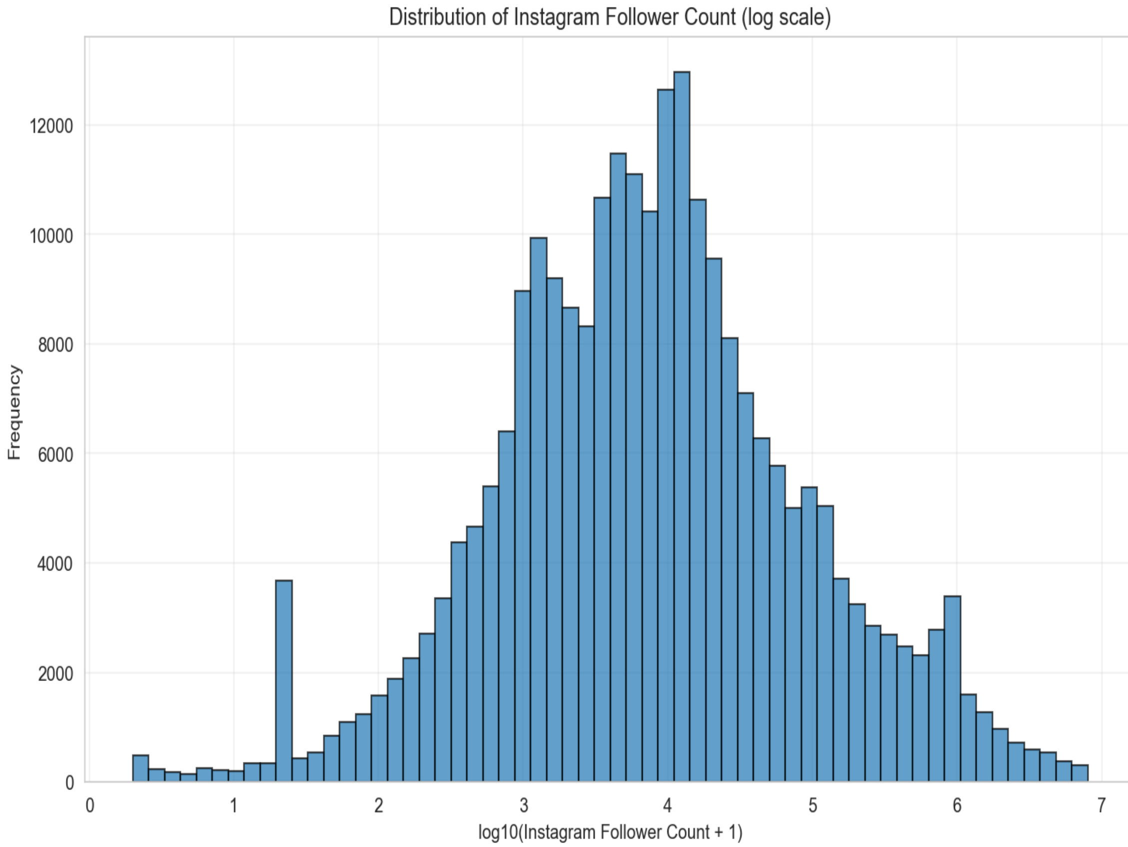
# Visualizations

### *Step 2: Correlation heatmap of key brand and influencer metrics.*



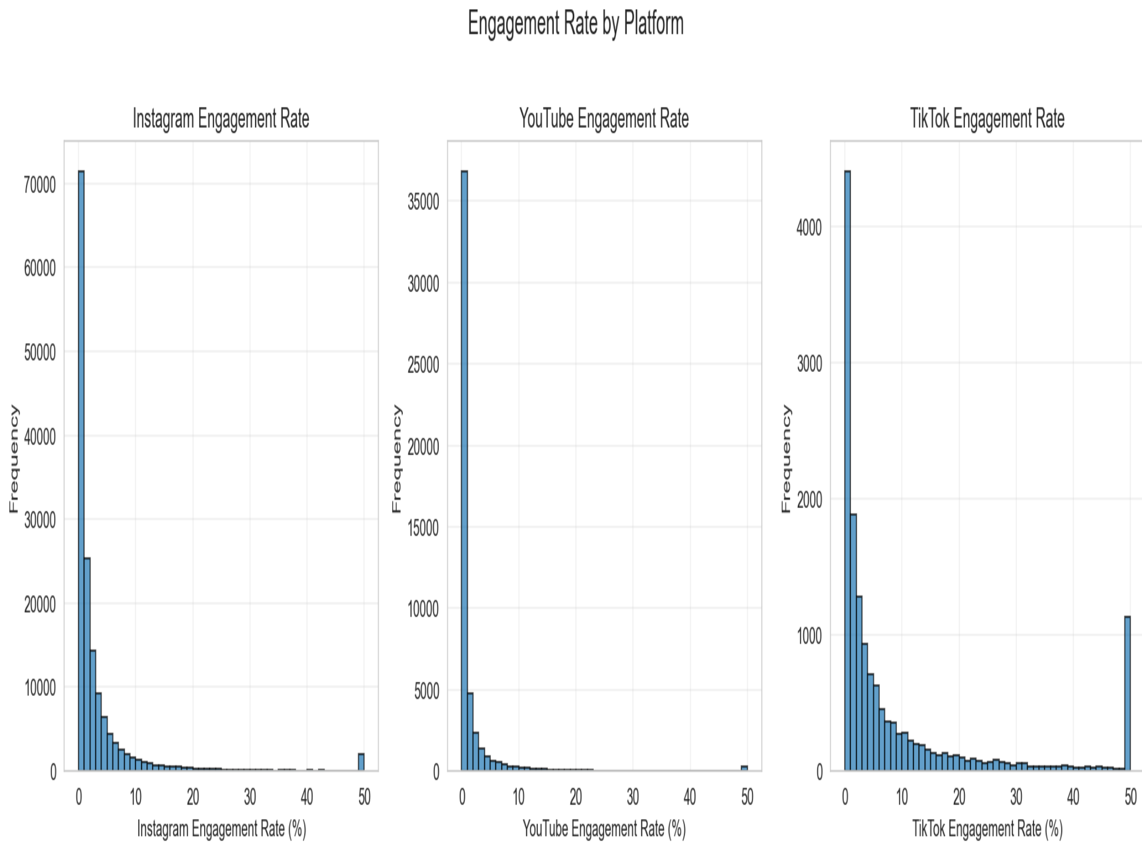

Correlation heatmap of key brand and influencer metrics.

### *Step 3: Distribution of Instagram follower count (log scale).*



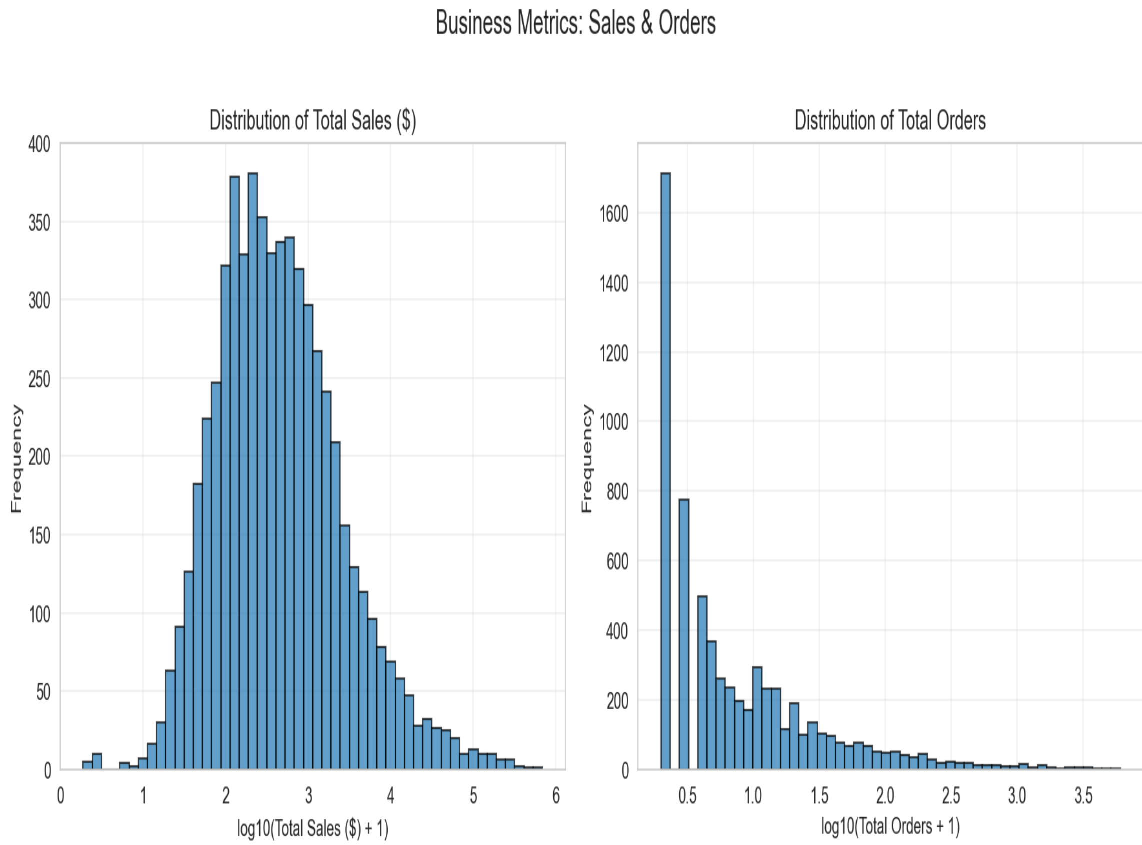

Distribution of Instagram follower count (log scale).

***Step 4: Engagement rate distribution by platform (Instagram, YouTube, TikTok).***



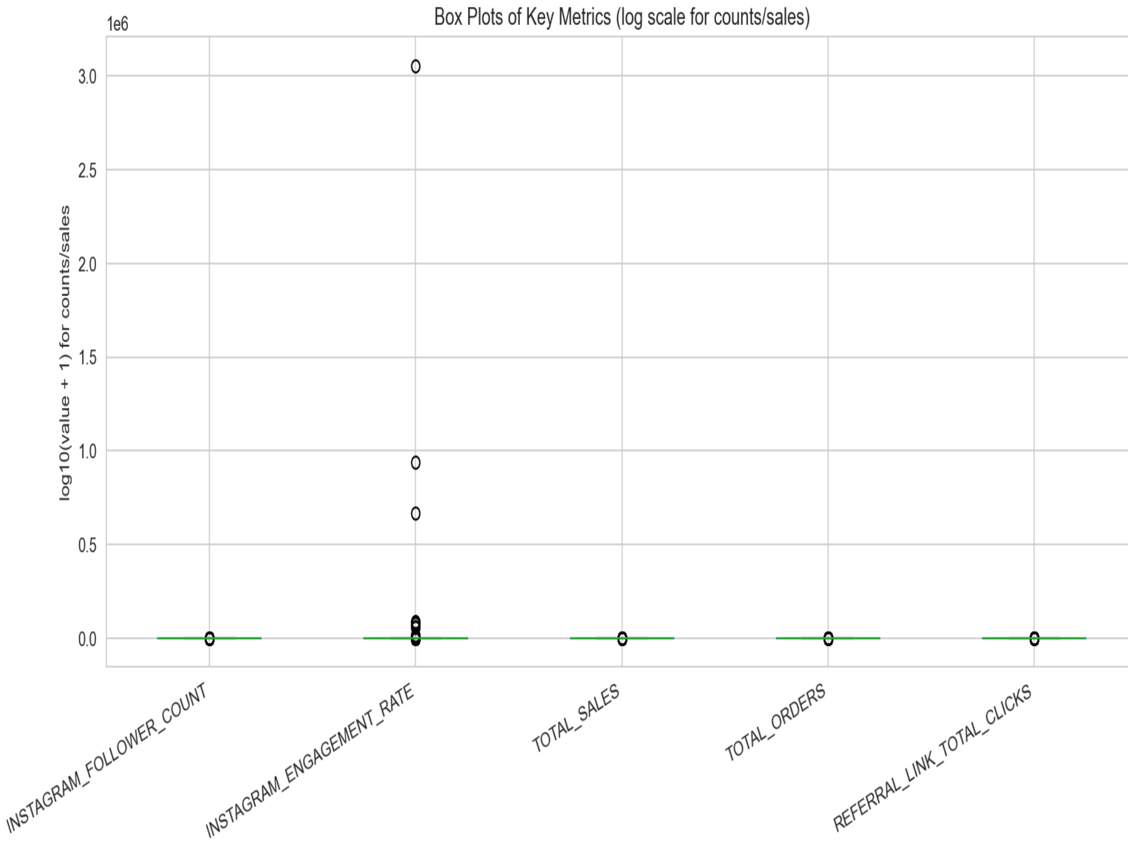

Engagement rate distribution by platform (Instagram, YouTube, TikTok).

***Step 5: Distribution of total sales and total orders (log scale).***



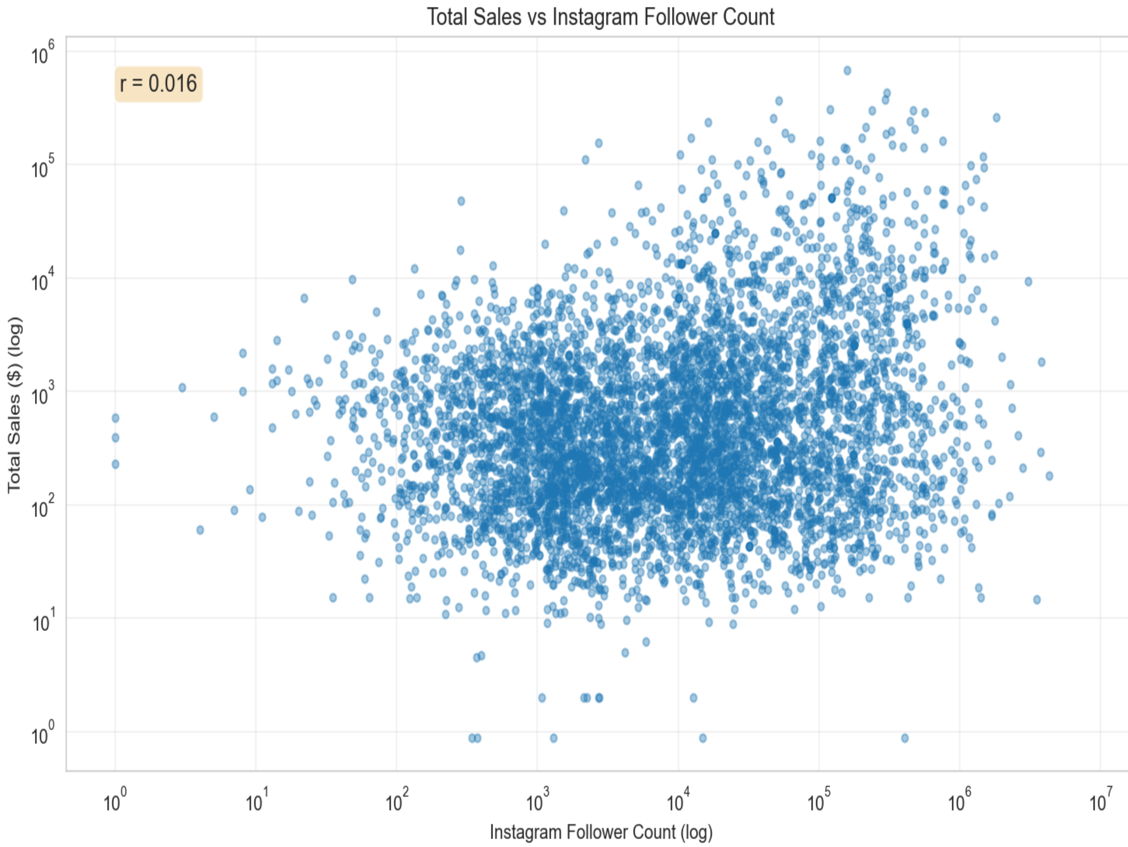

Distribution of total sales and total orders (log scale).

***Step 6: Box plots of key metrics to visualize spread and outliers.***



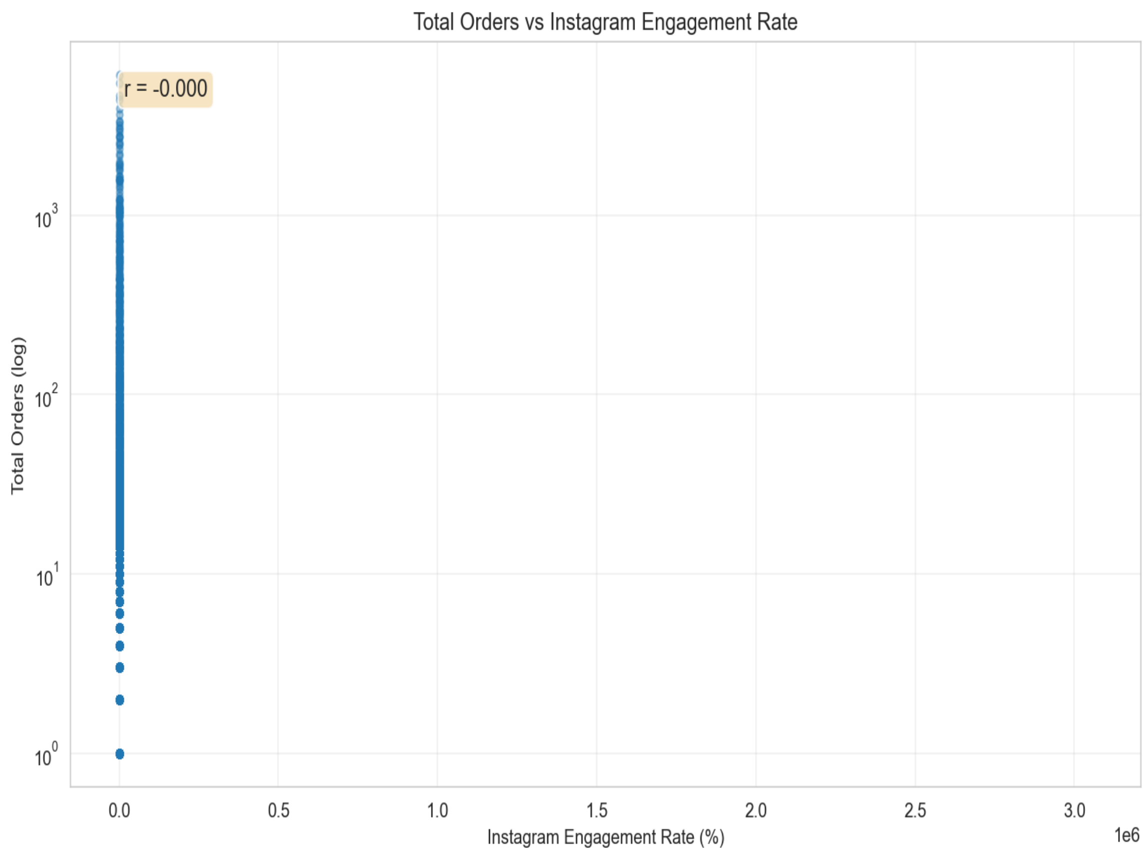

Box plots of key metrics to visualize spread and outliers.

### *Step 7: Scatter: Instagram followers vs total sales (log–log).*



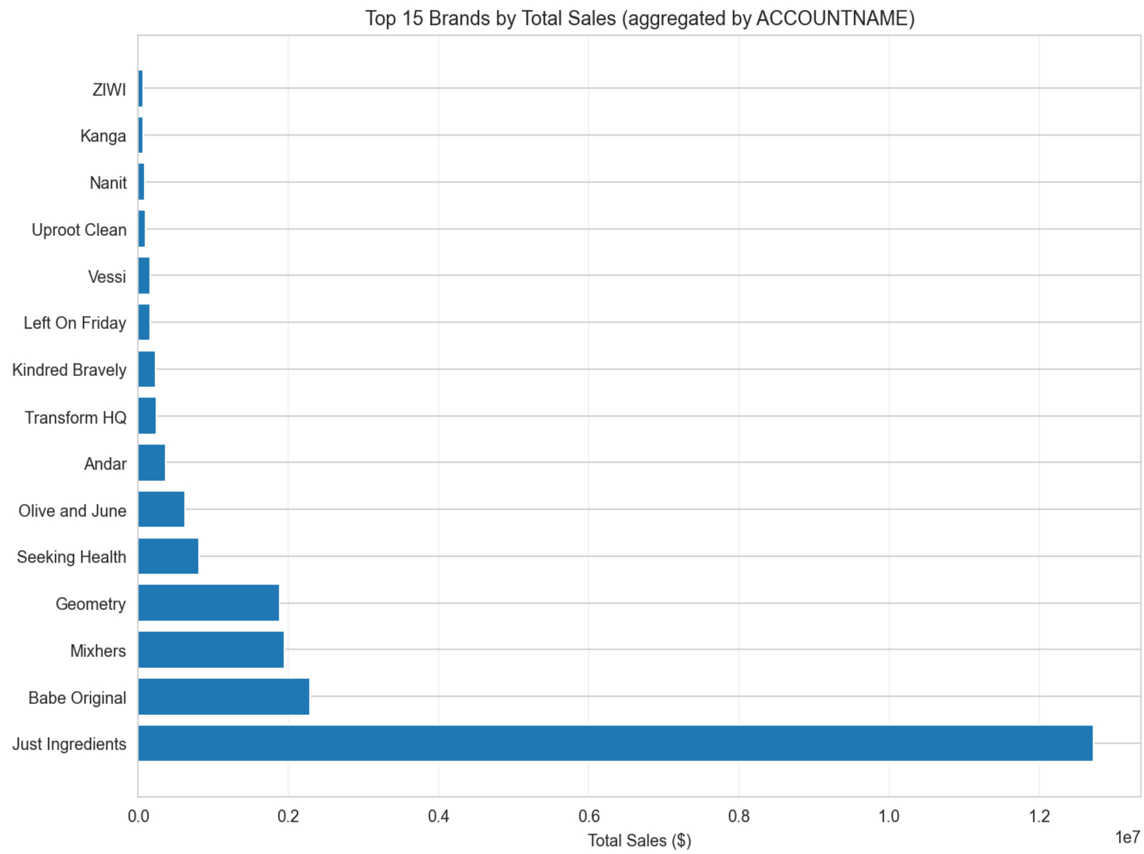

Scatter: Instagram followers vs total sales (log–log).

***Step 8: Scatter: Instagram engagement rate vs total orders.***



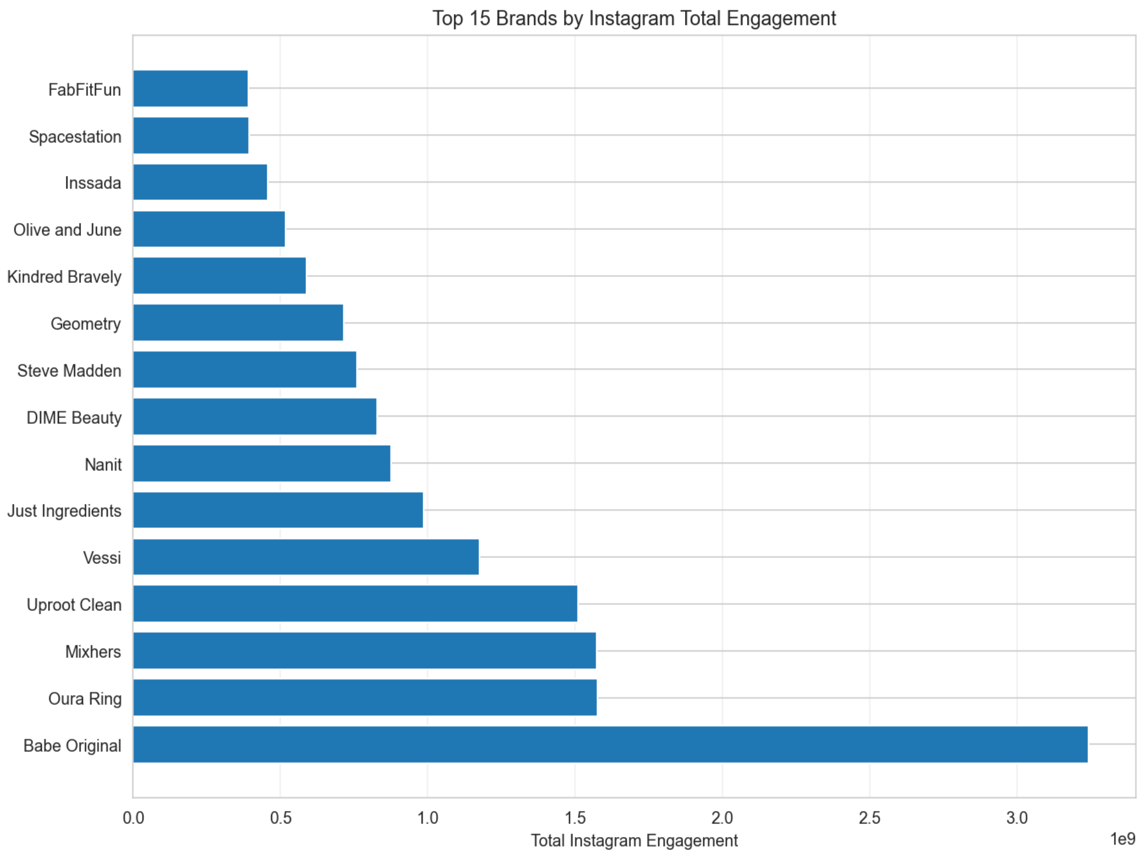
Scatter: Instagram engagement rate vs total orders.

### *Step 9: Top 15 brands by aggregated total sales.*



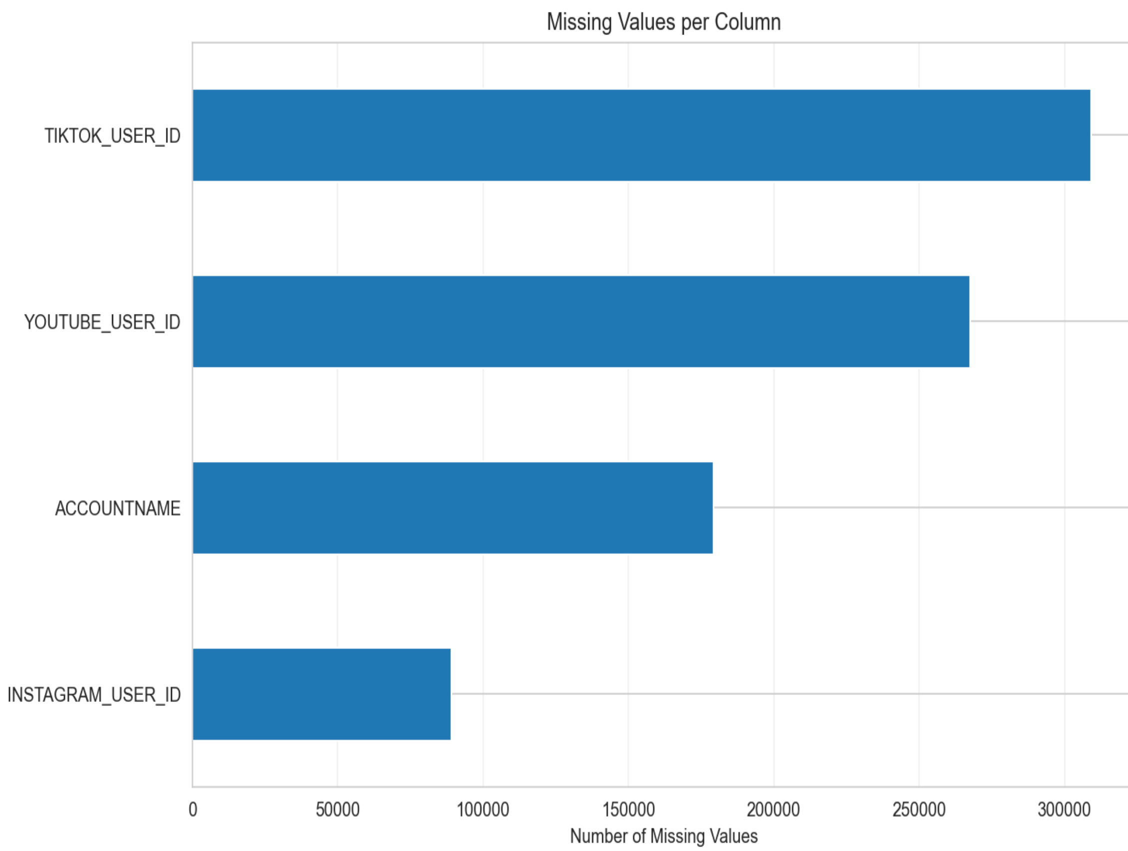
Top 15 brands by aggregated total sales.

***Step 10: Top 15 brands by Instagram total engagement.***



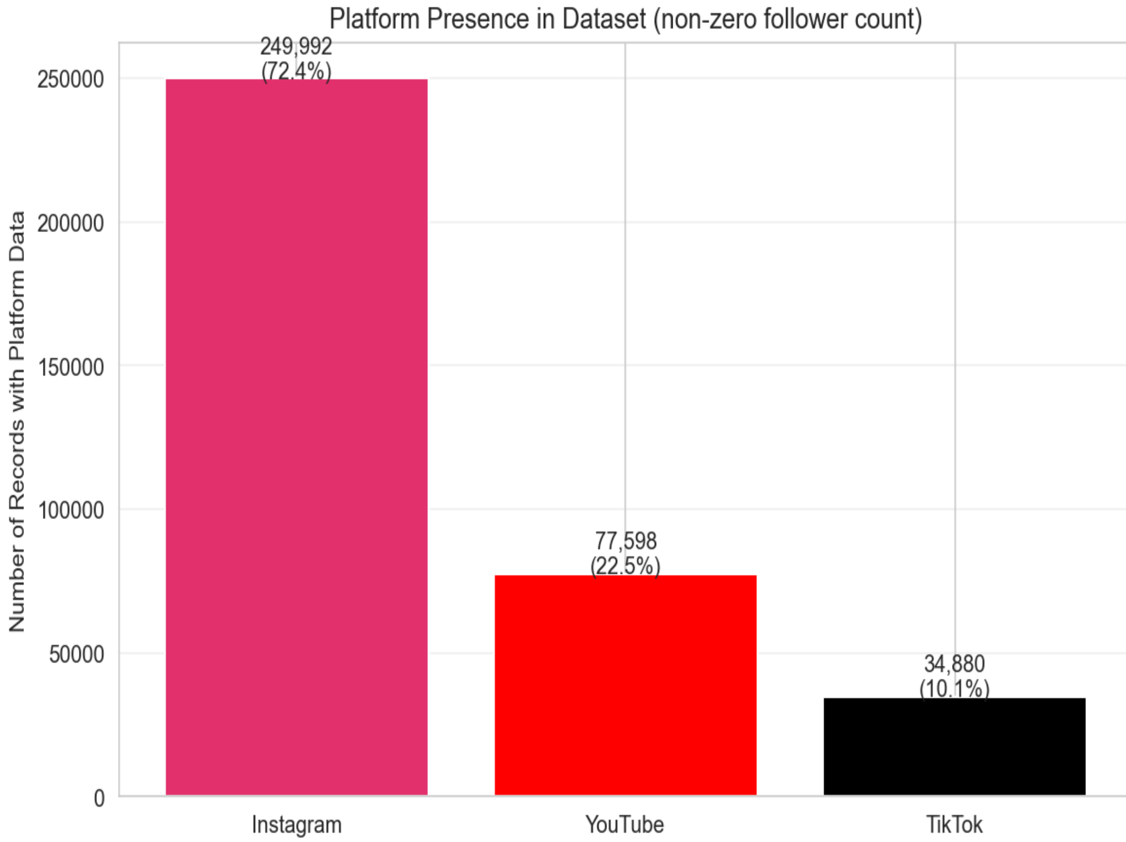

Top 15 brands by Instagram total engagement.

***Step 11: Missing values per column.***



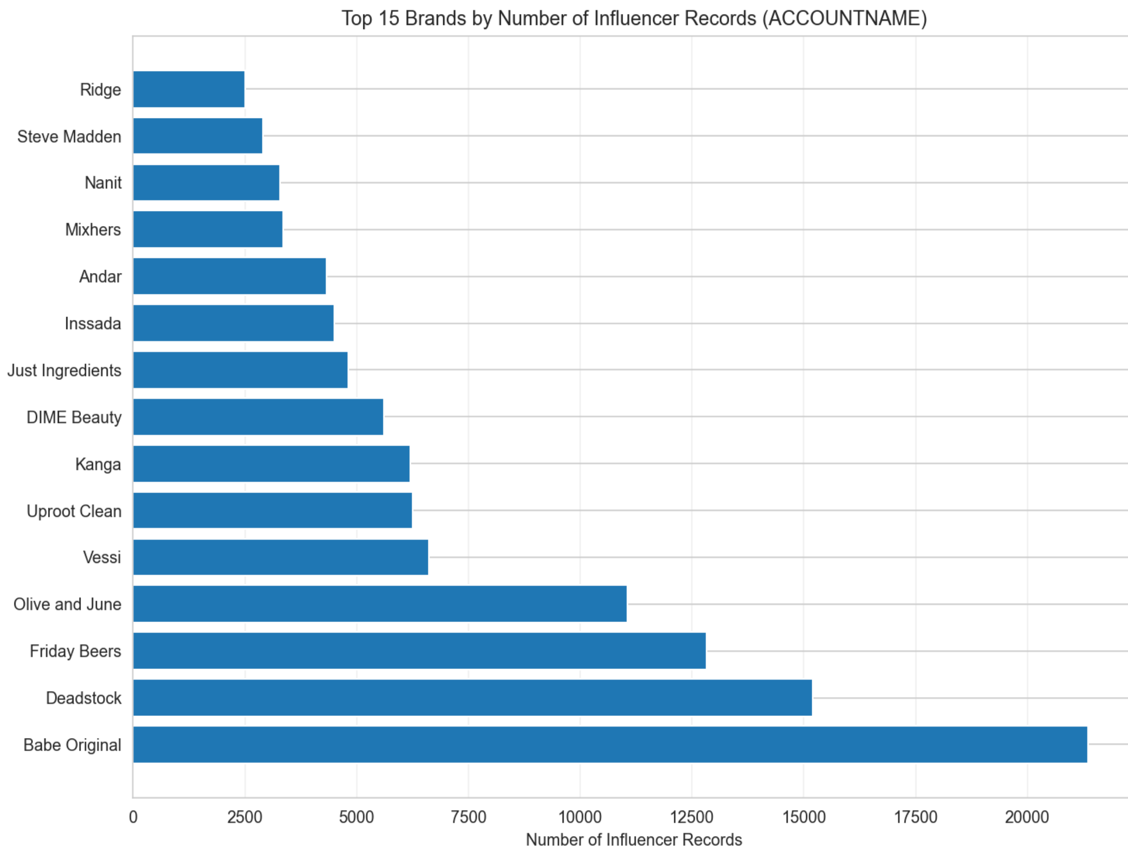
Missing values per column.

***Step 12: Platform presence: count of records with non-zero followers per platform.***



Platform presence: count of records with non-zero followers per platform.

***Step 13: Top 15 brands by number of influencer records.***




Top 15 brands by number of influencer records.

## Summary

The EDA covered data loading, missing value and correlation analysis, distributions of followers and engagement rates by platform, business metrics (sales and orders), outlier detection via box plots, and relationships between reach/engagement and conversions. Top brands were identified by total sales, Instagram engagement, and number of influencers. Platform presence and missing-value patterns were summarized to support downstream modeling and reporting.